BSC-MD: Developing an Android App

Proposal Sheet

Name: Stukalov Mikhail Vadimovich

Student ID: 3137193

Idea:

I want to build Restock, an app for keeping track of food and everyday supplies at home. Users can mark products as “In stock”, “Running low” or “Out of stock”. Items that are low or missing will appear on a shopping list. After buying an item, the user can mark it as restocked.

As an extension, I plan to add optional expiry dates and local reminders, for example two days before the saved date. A “Use soon” section will help users see which products to use first. Products with different expiry dates can be saved as separate entries.

Planned Activities/Layouts:

Home: a list of products grouped by category, with buttons to change their stock status.

Shopping list: low and missing products, manually added items, and a way to mark purchases as restocked.

Product details: add or edit a product’s name, category, status and optional expiry date. Products can also be deleted here.

Settings: turn the shake shortcut on or off and set reminder preferences if the expiry feature is added.

An explicit intent will open the product details activity from Home. An implicit intent will let users share their shopping list through another app.

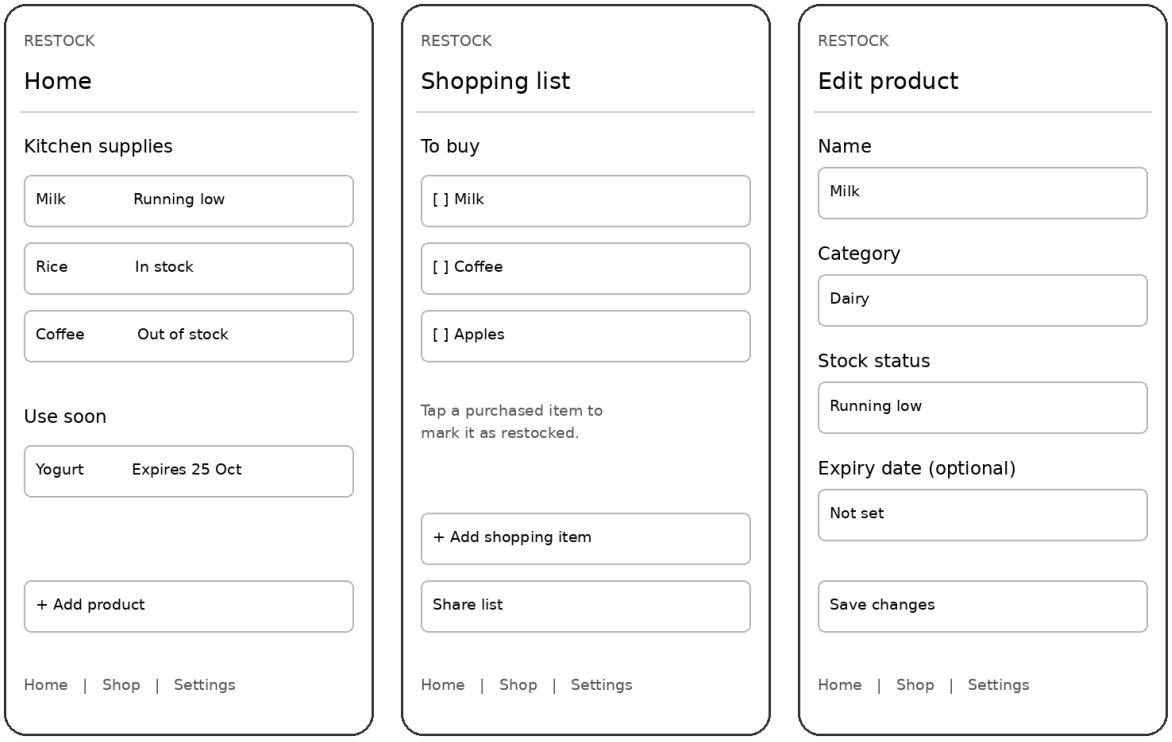
Sensor use:

The accelerometer will detect a deliberate shake while the app is open and switch to the shopping list. This gives users a quick way to check what they need to buy. A normal navigation button will also be available, and the gesture can be disabled in Settings.

Storage:

A local Room database will store product names, categories, stock statuses and shopping entries, plus expiry dates if the extension is added. Users will be able to create, view, update and delete records (CRUD). Marking a product as restocked will update its status and remove it from the automatic shopping list. The main features will work offline, without an account or server.

Mockups:


RESTOCK
Home
Kitchen supplies
Milk Running low
Rice In stock
Coffee Out of stock
Use soon
Yogurt Expires 25 Oct
+ Add product
Home | Shop | Settings
RESTOCK
Shopping list
To buy
[ ] Milk
[ ] Coffee
[ ] Apples
Tap a purchased item to mark it as restocked.
+ Add shopping item
Share list
Home | Shop | Settings
RESTOCK
Edit product
Name
Milk
Category
Dairy
Stock status
Running low
Expiry date (optional)
Not set
Save changes
Home | Shop | Settings